Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 - Machine Learning Algorithms Laboratory		
Academic year	2025-2026 (Even)	Batch: 2023-2027	**Due date: 06/01/2026**

Experiment 1: Working with Python Packages – Numpy, Scipy, Scikit-Learn, Matplotlib

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Aim:
The goal of this experiment is to get comfortable with the core Python libraries like Numpy, Pandas, Scikit-learn, and Matplotlib. I focused on loading datasets from public repositories, performing EDA (Exploratory Data Analysis) to clean and understand the data, and then implementing the correct Machine Learning models (Regression vs. Classification) based on the problem type.

Libraries Used:

- **Pandas & Numpy:** I used these for loading the CSV files and handling array operations.
- **Matplotlib & Seaborn:** Used to create the visualizations (boxplots, heatmaps, and distribution plots) to check for outliers and data spread.
- **Scikit-Learn:**
  - *Preprocessing:* StandardScaler (for scaling features) and LabelEncoder (for categorical data).
  - *Selection:* SelectKBest and Chi2 to find the most relevant features.
  - *Models:* Random Forest, SVM, Naive Bayes, and KNN.
- **Math & CV2**

Mathematical Concepts:

- **Why EDA?** Before feeding data to a model, it is important to understand its structure and behavior. Exploratory Data Analysis helps identify data distribution, missing values, outliers, and relationships between features.
- **Summary Statistics:** Basic statistical measures such as Mean and Variance were used to understand the spread of data.

$$\mu = \frac{1}{n}\sum x_i \qquad s^2 = \frac{1}{n-1}\sum (x_i - \mu)^2$$

- **Finding Outliers:** Boxplots were used for outlier detection based on the Interquartile Range (IQR).

$$IQR = Q_3 - Q_1$$

Any data point outside the range

$$Q_1 - 1.5 \times IQR \quad \text{or} \quad Q_3 + 1.5 \times IQR$$

was considered an outlier.

- **Feature Correlation:** To identify relationships between numerical features, the Pearson Correlation Coefficient was used:

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

  Values close to $+1$ or -1 indicate strong positive or negative correlation.

- **Feature Scaling:** Since features existed on different numeric scales, standardization was applied to bring them to a common scale, improving model performance.

- **Feature Selection:** Statistical techniques such as Chi-Square test and ANOVA were used to select the most relevant features for classification tasks.

Results and Observations:

I applied the same workflow (Load → Clean → Visualize → Model) across five different datasets. Here are the observations for each.

0.1 1. Loan Amount Prediction (Regression)

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14
person_age	person_gender	person_education	person_income	person_emp_exp	person_home_ownership	loan_amnt	loan_intent	loan_int_rate	loan_percent_income	cb_person_cred_hist_length	credit_score	previous_loan_defaults_on_file	loan_status
22.0	female	Master	71948.0	0	RENT	35000.0	PERSONAL	16.02	0.49	3.0	561	No	1
21.0	female	High School	12282.0	0	OWN	1000.0	EDUCATION	11.14	0.08	2.0	504	Yes	0
25.0	female	High School	12438.0	3	MORTGAGE	5500.0	MEDICAL	12.87	0.44	3.0	635	No	1
23.0	female	Bachelor	79753.0	0	RENT	35000.0	MEDICAL	15.23	0.44	2.0	675	No	1
24.0	male	Master	66135.0	1	RENT	35000.0	MEDICAL	14.27	0.53	4.0	586	No	1
21.0	female	High School	12951.0	0	OWN	2500.0	VENTURE	7.14	0.19	2.0	532	No	1
26.0	female	Bachelor	93471.0	1	RENT	35000.0	EDUCATION	12.42	0.37	3.0	701	No	1
24.0	female	High School	95550.0	5	RENT	35000.0	MEDICAL	11.11	0.37	4.0	585	No	1
24.0	female	Associate	100684.0	3	RENT	35000.0	PERSONAL	8.9	0.35	2.0	544	No	1
21.0	female	High School	12739.0	0	OWN	1600.0	VENTURE	14.74	0.13	3.0	640	No	1
22.0	female	High School	102985.0	0	RENT	35000.0	VENTURE	10.37	0.34	4.0	621	No	1
21.0	female	Associate	13113.0	0	OWN	4500.0	HOMEIMPROVEMENT	8.63	0.34	2.0	651	No	1
23.0	male	Bachelor	114860.0	3	RENT	35000.0	VENTURE	7.9	0.3	2.0	573	No	1
26.0	male	Master	130713.0	0	RENT	35000.0	EDUCATION	18.39	0.27	4.0	708	No	1
23.0	female	Associate	138998.0	0	RENT	35000.0	EDUCATION	7.9	0.25	4.0	583	No	0
23.0	female	Master	600891.0	5	MORTGAGE	30000.0	DEBTCONSOLIDATION	10.65	0.05	3.0	670	Yes	0
23.0	male	Bachelor	144943.0	0	RENT	35000.0	EDUCATION	7.9	0.24	4.0	663	No	0
23.0	female	High School	111369.0	0	RENT	35000.0	MEDICAL	20.0	0.31	4.0	694	No	1
23.0	male	Bachelor	136628.0	0	RENT	35000.0	DEBTCONSOLIDATION	18.25	0.26	4.0	709	No	1
24.0	female	Master	14283.0	1	MORTGAGE	1750.0	EDUCATION	10.99	0.12	2.0	679	No	1
25.0	male	Bachelor	195718.0	0	RENT	35000.0	VENTURE	7.49	0.18	4.0	684	Yes	0
25.0	male	High School	165792.0	4	RENT	34800.0	PERSONAL	16.77	0.21	2.0	662	No	0

Figure 1: Loan Data: Checking column types

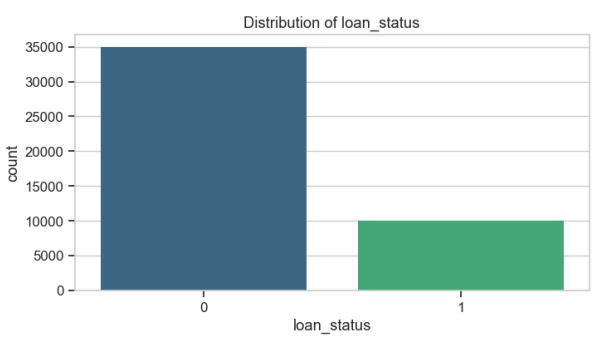

Figure 2: Distribution of Loan Amounts

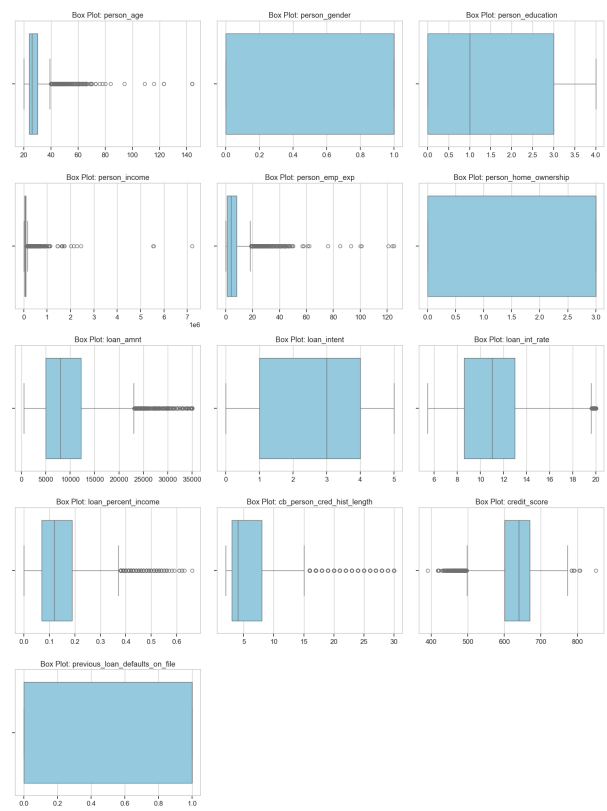

Figure 3: Boxplot showing outliers

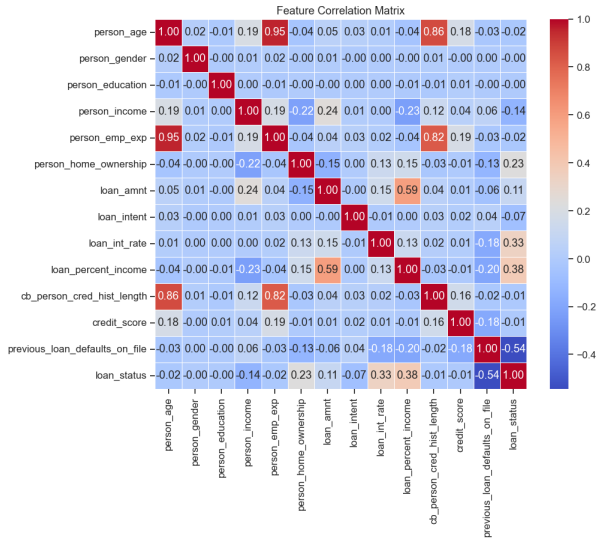

Figure 4: Correlation Heatmap

```
Classification Report:
              precision    recall  f1-score   support

           0       0.94      0.97      0.95      6990
           1       0.89      0.77      0.83      2010

    accuracy                           0.93      9000
   macro avg       0.91      0.87      0.89      9000
weighted avg       0.93      0.93      0.93      9000
```

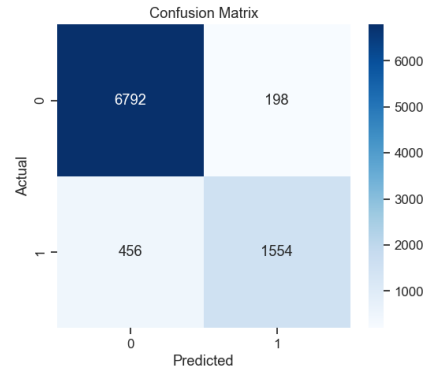

Figure 5: Model Performance Report

0.2 2. Predicting Diabetes (Binary Classification)

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
1	Female	80.0	0	1	never	25.19	6.6	140	0
2	Female	54.0	0	0	No Info	27.32	6.6	80	0
3	Male	28.0	0	0	never	27.32	5.7	158	0
4	Female	36.0	0	0	current	23.45	5.0	155	0
5	Male	76.0	1	1	current	20.14	4.8	155	0
6	Female	20.0	0	0	never	27.32	6.6	85	0
7	Female	44.0	0	0	never	19.31	6.5	200	1
8	Female	79.0	0	0	No Info	23.86	5.7	85	0
9	Male	42.0	0	0	never	33.64	4.8	145	0
10	Female	32.0	0	0	never	27.32	5.0	100	0

Figure 6: Diabetes Data Overview

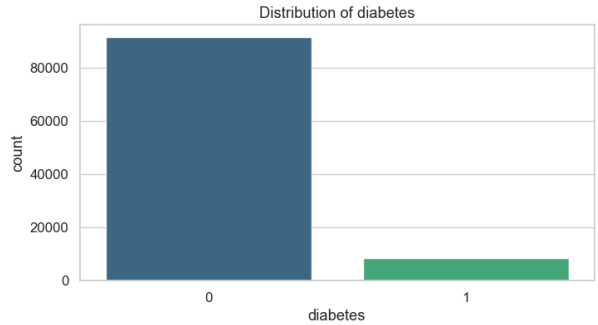

Figure 7: Data Distribution

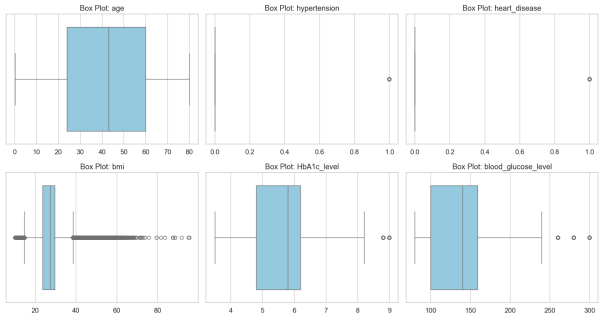

Figure 8: Outlier Detection

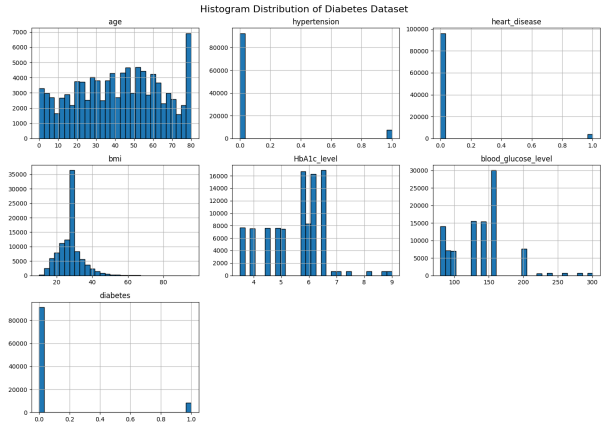

Figure 9: Feature Histograms

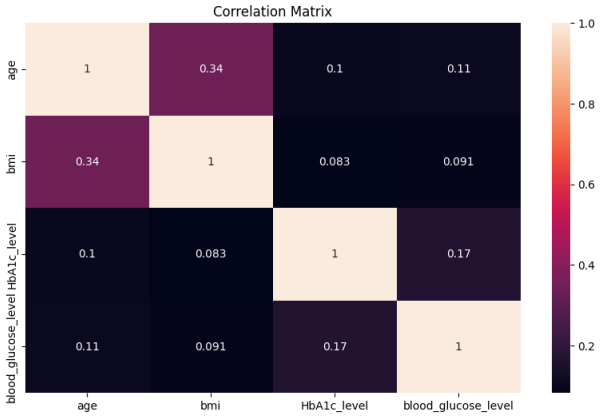

Figure 10: Correlation Matrix

```
Model Performance
Accuracy: 0.9587

Classification Report:
              precision    recall  f1-score   support

           0       0.96      1.00      0.98     18292
           1       0.92      0.56      0.70      1708

    accuracy                           0.96     20000
   macro avg       0.94      0.78      0.84     20000
weighted avg       0.96      0.96      0.95     20000
```

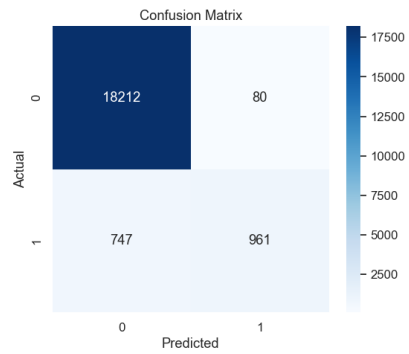

Figure 11: Classification Report

0.3 3. Email Spam Classification (NLP)

Observation: For this task, I treated the email text as high-dimensional data using a Bag-of-Words representation. I noticed during EDA that the class balance between spam and ham was slightly skewed, which is typical for this type of problem. By plotting the word count distribution, I saw that spam emails often have different length patterns. I applied normalization to ensure the model wouldn't be biased by the length of the emails.

	<anonymous>	label	text	label_num
1	605	ham	Subject: enron methanol ; meter # : 988291⏎this is a foll	0
2	2349	ham	Subject: hpl nom for january 9 , 2001⏎(see attached file	0
3	3624	ham	Subject: neon retreat⏎ho ho ho , we ' re around to that m	0
4	4685	spam	Subject: photoshop , windows , office . cheap . main tren	1
5	2030	ham	Subject: re : indian springs⏎this deal is to book the tec	0
6	2949	ham	Subject: ehronline web address change⏎this message is int	0
7	2793	ham	Subject: spring savings certificate - take 30 % off⏎save	0
8	4185	spam	Subject: looking for medication ? we ` re the best source	1
9	2641	ham	Subject: noms / actual flow for 2 / 26⏎we agree⏎- - - - -	0
10	1870	ham	Subject: nominations for oct . 21 - 23 , 2000⏎(see attac	0

Figure 12: Spam Data: Feature matrix columns

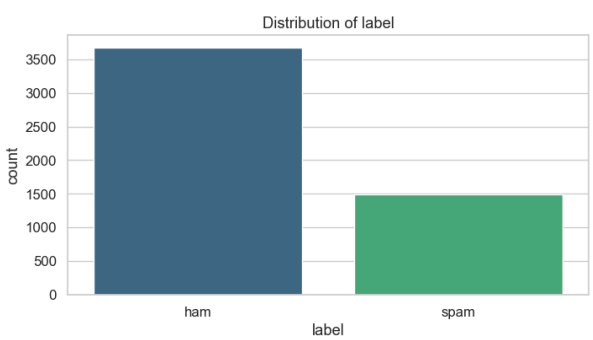

Figure 13: Spam vs Ham Class Distribution

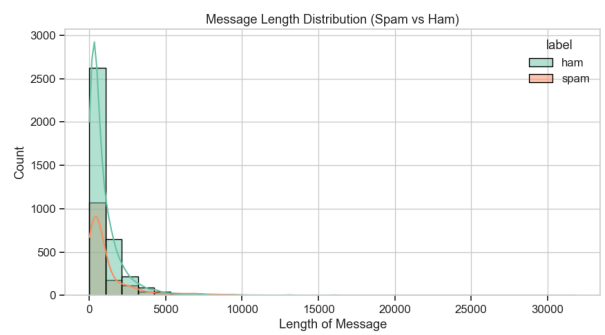

Figure 14: Spam-Ham Message Length Distribution

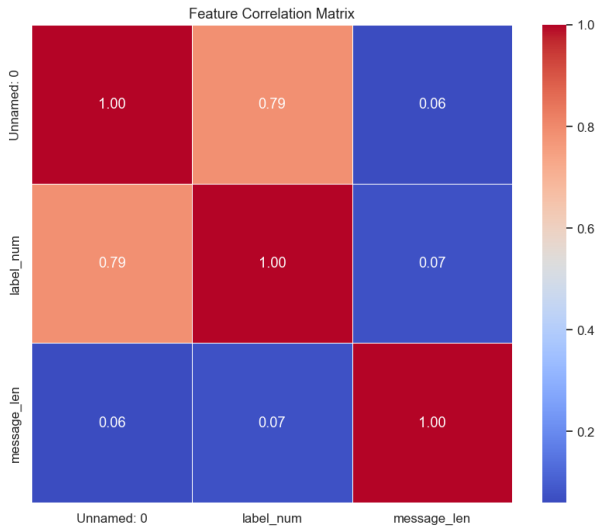

Figure 15: Feature Correlation Matrix

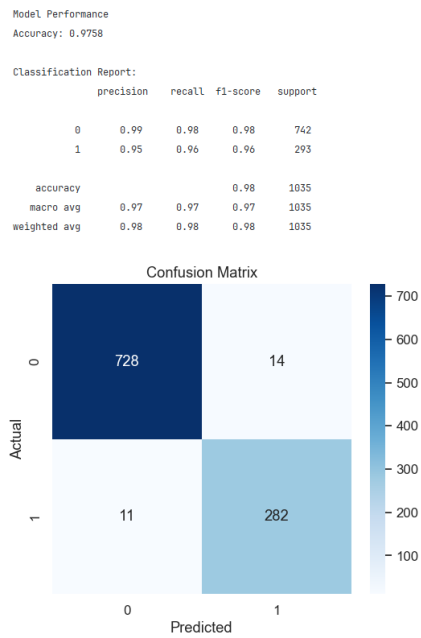

Figure 16: Spam Classification Report

0.4 4. Handwritten Character Recognition (MNIST)

Observation: This dataset deals with 8x8 pixel images. The pixel intensity distribution showed a lot of zeros (black background). Since the relationships between pixels are non-linear, I chose a Random Forest Classifier.

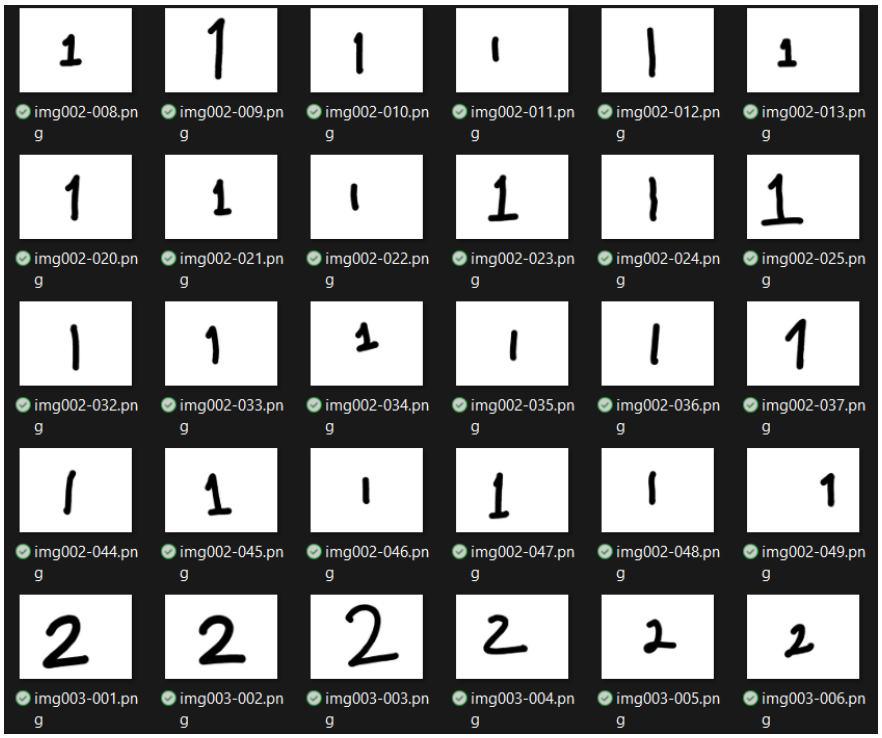

Figure 17: Sample Digits from dataset

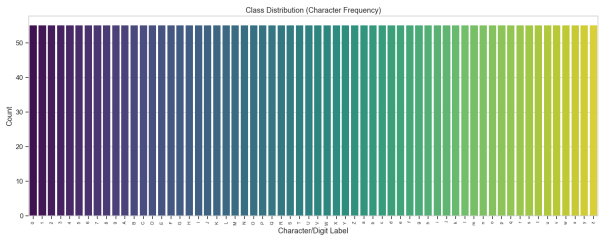

Figure 18: Class Balance

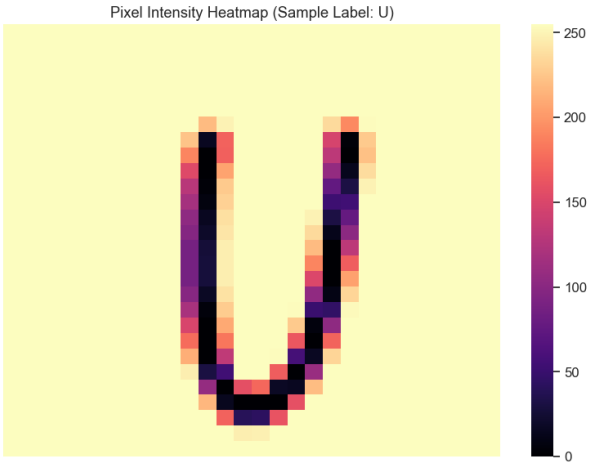

Figure 19: Pixel Intensity Spread

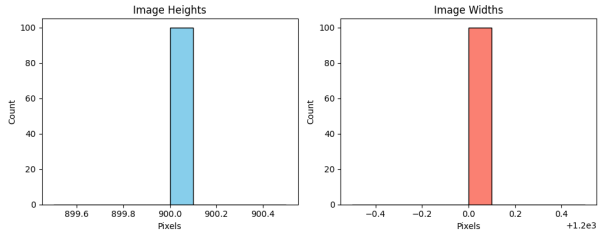

Figure 20: Height vs Weight Analysis

accuracy			0.44	682
macro avg	0.46	0.45	0.44	682
weighted avg	0.49	0.44	0.45	682

Figure 21: MNIST Report

0.5 5. Iris Dataset (Multi-class Classification)

Observation: A simple multi-class problem. The combined plots clearly showed that Petal Length and Width are the best features to separate the species, which makes KNN a perfect candidate for this task.

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	1	5.1	3.5	1.4	0.2	Iris-setosa
2	2	4.9	3.0	1.4	0.2	Iris-setosa
3	3	4.7	3.2	1.3	0.2	Iris-setosa
4	4	4.6	3.1	1.5	0.2	Iris-setosa
5	5	5.0	3.6	1.4	0.2	Iris-setosa
6	6	5.4	3.9	1.7	0.4	Iris-setosa
7	7	4.6	3.4	1.4	0.3	Iris-setosa
8	8	5.0	3.4	1.5	0.2	Iris-setosa

Figure 22: Iris Data Structure

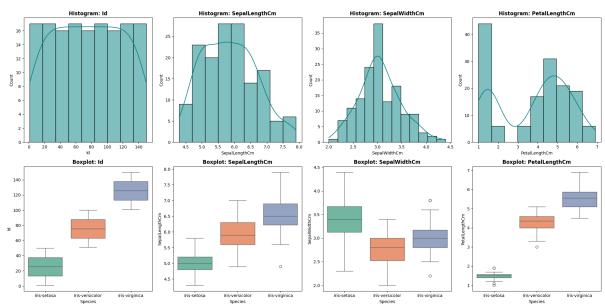

Figure 23: Combined Hist & Boxplot

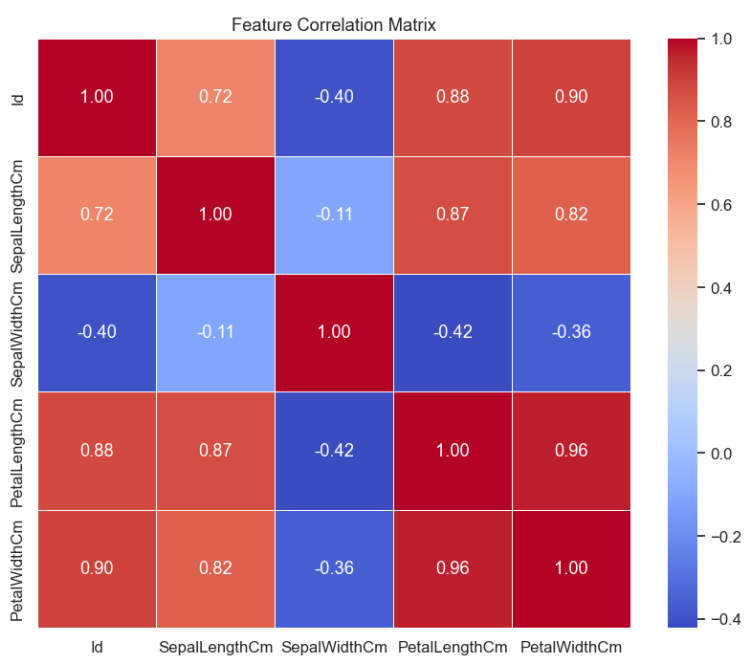

Figure 24: Feature Correlation Matrix

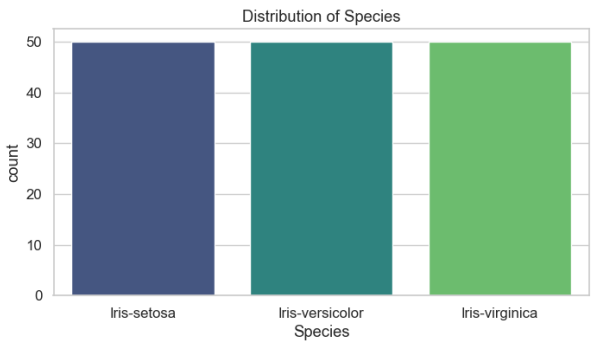

Figure 25: Class Balance

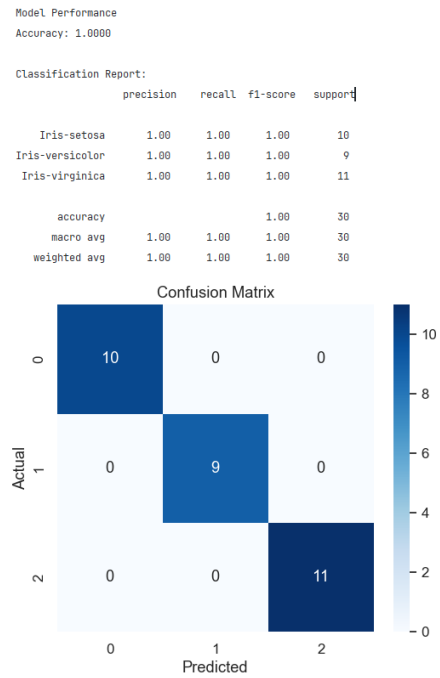

Figure 26: Iris Dataset Report

Final Inference Table:

Dataset	**Task Type**	**Feature Selection**	**Algorithm Used**
Iris Dataset	Classification	ANOVA (F-Classif)	K-Nearest Neighbors (KNN)
Predicting Diabetes	Classification	Chi-Square Test	Support Vector Machine (SVM)
Email Spam	Classification	Bag of Words	Naive Bayes
MNIST Digits	Classification	Raw Pixels / PCA	Random Forest Classifier
Loan Prediction	Regression	Correlation	Random Forest Regressor

Learning Outcomes:

- **Data Handling:** I learned how to properly inspect a raw dataset, use heatmaps to find missing data, and fix data types using Pandas.
- **Visualization:** I got comfortable using Histograms to check distributions and Boxplots to quickly spot statistical outliers ($1.5 \times IQR$).
- **Feature Selection:** I understood that we have to use Correlation Matrices and tests like ANOVA to pick only the useful features.